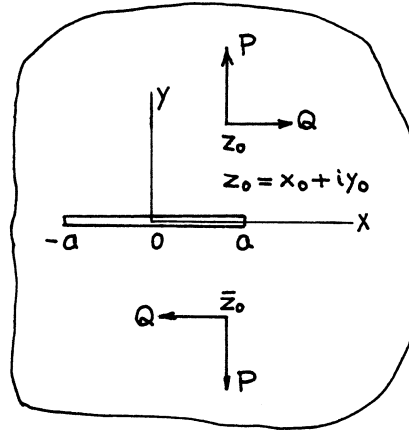




$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \frac{1}{2} \left(\frac{\sqrt{a^2 - z_0^2}}{z - z_0} + \frac{\sqrt{a^2 - \bar{z}_0^2}}{z - \bar{z}_0} \right) \frac{1}{\sqrt{z^2 - a^2}}$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \left\{ \tan^{-1} \sqrt{\frac{a+z_0}{a-z_0}} \sqrt{\frac{z-a}{z+a}} + \tan^{-1} \sqrt{\frac{a+\bar{z}_0}{a-\bar{z}_0}} \sqrt{\frac{z-a}{z+a}} \right\}$$

$$K_{I\pm a} = \frac{P}{\sqrt{\pi a}} \left[1 - \alpha y_0 \frac{\partial}{\partial y_0} \right] \frac{1}{2} \left\{ \sqrt{\frac{a \pm z_0}{a \mp z_0}} + \sqrt{\frac{a \pm \bar{z}_0}{a \mp \bar{z}_0}} \right\}$$

$$K_{II\pm a} = \frac{Q}{\sqrt{\pi a}} \left[1 + \alpha y_0 \frac{\partial}{\partial y_0} \right] \frac{1}{2} \left\{ \sqrt{\frac{a \pm z_0}{a \mp z_0}} + \sqrt{\frac{a \pm \bar{z}_0}{a \mp \bar{z}_0}} \right\}$$

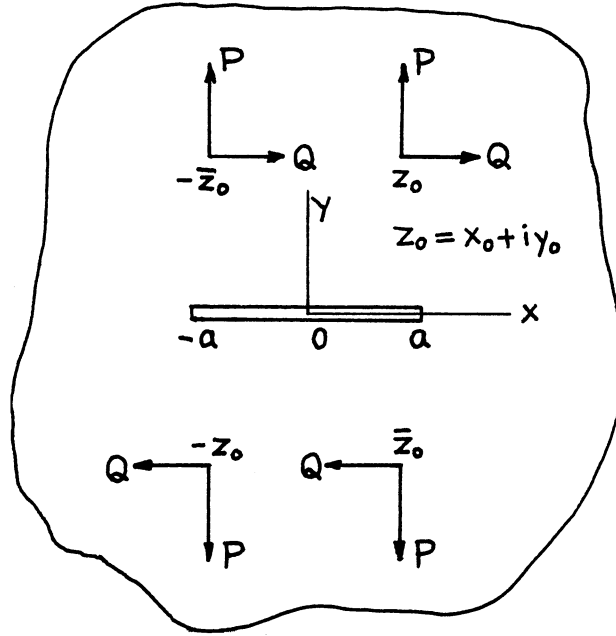
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function

Accuracy: Exact

References: **Tada 1972a, 1973**



$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \left(\frac{\sqrt{a^2 - z_0^2}}{z^2 - z_0^2} + \frac{\sqrt{a^2 - \bar{z}_0^2}}{z^2 - \bar{z}_0^2} \right) \frac{1}{\sqrt{1 - (a/z)^2}}$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \left\{ \tan^{-1} \sqrt{\frac{z^2 - a^2}{a^2 - z_0^2}} + \tan^{-1} \sqrt{\frac{z^2 - a^2}{a^2 - \bar{z}_0^2}} \right\}$$

$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \left\{ \frac{a}{\sqrt{a^2 - z_0^2}} + \frac{a}{\sqrt{a^2 - \bar{z}_0^2}} \right\}$$

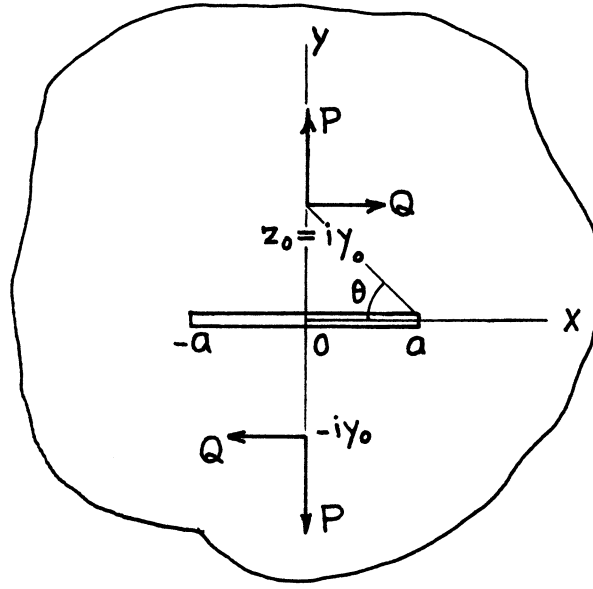
where

$$\alpha = \begin{cases} \frac{1}{2} (1 + \nu) & \text{plane stress} \\ \frac{1}{2} \left(\frac{1}{1 - \nu} \right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function (Superposition of **page 5.6**)

Accuracy: Exact

References: **Tada 1972a, 1973**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \frac{\sqrt{a^2 + y_0^2}}{z^2 + y_0^2} \frac{1}{\sqrt{1 - (a/z)^2}}$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \tan^{-1} \sqrt{\frac{z^2 - a^2}{a^2 + y_0^2}}$$

$$\begin{aligned} \begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} &= \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \frac{a}{\sqrt{a^2 + y_0^2}} \\ &= \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \end{Bmatrix} \frac{a}{\sqrt{a^2 + y_0^2}} \left[1 \begin{Bmatrix} + \\ - \end{Bmatrix} \alpha \frac{y_0^2}{a^2 + y_0^2} \right] \left(= \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \end{Bmatrix} \cos \theta \left[1 \begin{Bmatrix} + \\ - \end{Bmatrix} \alpha \sin^2 \theta \right] \right) \end{aligned}$$

$$\begin{aligned} \text{Im} \begin{Bmatrix} \bar{Z}_I(x) \\ \bar{Z}_{II}(x) \end{Bmatrix} &= \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \tanh^{-1} \sqrt{\frac{a^2 - x^2}{a^2 + y_0^2}} \\ &= \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[\tanh^{-1} \sqrt{\frac{a^2 - x^2}{a^2 + y_0^2}} \begin{Bmatrix} + \\ - \end{Bmatrix} \alpha \frac{y_0^2}{x^2 + y_0^2} \sqrt{\frac{a^2 - x^2}{a^2 + y_0^2}} \right] \end{aligned}$$

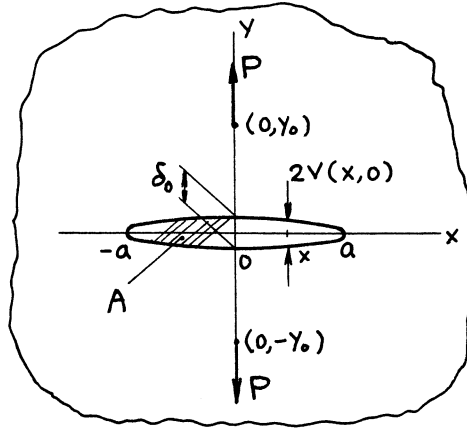
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function, etc. (Special Case of **page 5.6**)

Accuracy: Exact

References: **Paris 1957; Barenblatt 1962; Irwin 1962a; Tada 1970, 1973**



$$K_I = \frac{P}{\sqrt{\pi a}} \frac{a}{\sqrt{a^2 + y_0^2}} \left(1 + \alpha \frac{y_0^2}{a^2 + y_0^2} \right)$$

Crack Opening Area:

$$A = \frac{4P}{E'} \left(\sqrt{a^2 + y_0^2} - y_0 \right) \left(1 + \alpha \frac{y_0}{\sqrt{a^2 + y_0^2}} \right)$$

Crack Opening Profile:

$$2v(x, 0) = \frac{4P}{\pi E'} \left(\tanh^{-1} \sqrt{\frac{a^2 - x^2}{a^2 + y_0^2}} + \alpha \frac{y_0^2}{x^2 + y_0^2} \sqrt{\frac{a^2 - x^2}{a^2 + y_0^2}} \right)$$

Opening at Center:

$$\delta_0 = 2v(0, 0) = \frac{4P}{\pi E'} \left(\sinh^{-1} \frac{a}{y_0} + \alpha \frac{a}{\sqrt{a^2 + y_0^2}} \right)$$

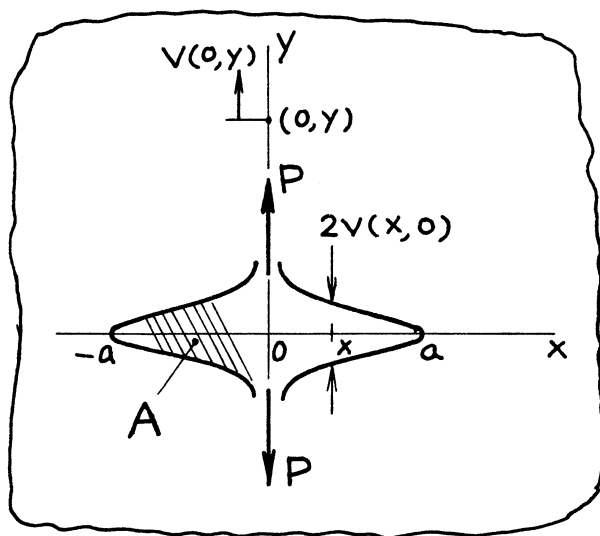
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Methods: A , ν Paris' Equation (see **Appendix B**) (or ν also from **page 5.8**)

Accuracy: Exact

Reference: **Tada 1985**



$$Z_I(z) = \frac{P}{\pi} \frac{a}{z\sqrt{z^2 - a^2}}$$

$$\overline{Z}_I(z) = \frac{P}{\pi} \cos^{-1} \frac{a}{z}$$

$$K_I = \frac{P}{\sqrt{\pi a}}$$

$$A = \frac{4Pa}{E'}$$
$$2v(x, 0) = \frac{4P}{\pi E'} \cosh^{-1} \frac{a}{x}$$

Vertical Displacement at $(0, y)$:

$$v(0, y) = \frac{2P}{\pi E'} \left(\sinh^{-1} \frac{a}{y} + \alpha \frac{a}{\sqrt{a^2 + y^2}} \right)$$

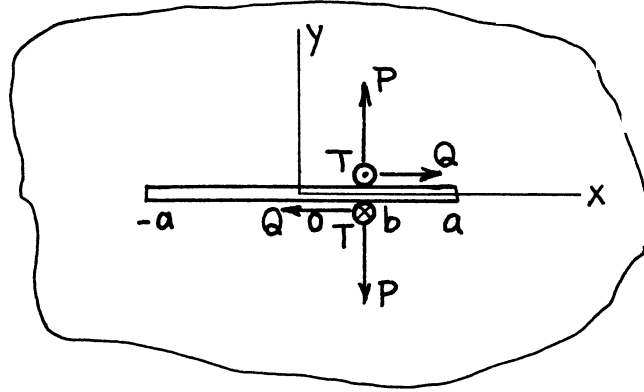
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function (Special Case of **page 5.8 or 5.10**)

Accuracy: Exact

Reference: **Tada 1985**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{\sqrt{a^2 - b^2}}{(z - b)\sqrt{z^2 - a^2}}$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \\ \bar{Z}_{III}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \sin^{-1} \frac{bz - a^2}{a(z - b)}$$

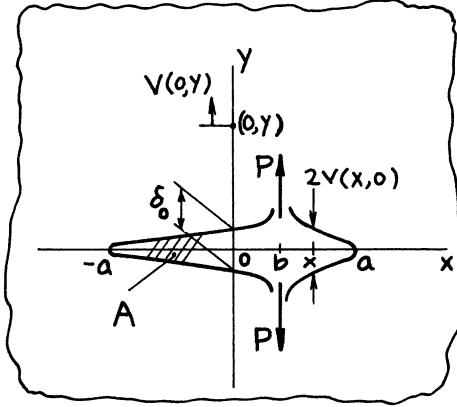
$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_{\pm a} = \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \sqrt{\frac{a \pm b}{a \mp b}}$$

$$\operatorname{Im} \begin{Bmatrix} \bar{Z}_I(x) \\ \bar{Z}_{II}(x) \\ \bar{Z}_{III}(x) \end{Bmatrix}_{|x| \leq a} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \cosh^{-1} \frac{a^2 - bx}{a|x - b|}$$

Method: Westergaard Stress Function, etc.

Accuracy: Exact

References: Irwin 1957, 1958a; Erdogan 1962; Sih 1962a, 1964; Paris 1965



$$K_{I\pm a} = \frac{P}{\sqrt{\pi a}} \frac{\sqrt{a^2 - b^2}}{a \mp b}$$

Crack Opening Area:

$$A = \frac{4P}{E'} \sqrt{a^2 - b^2}$$

Crack Opening Profile:

$$2v(x, 0) = \frac{4P}{\pi E'} \cosh^{-1} \frac{a^2 - bx}{a|x - b|} \quad |x| \leq a$$

Opening at Center:

$$\delta_0 = 2v(0, 0) = \frac{4P}{\pi E'} \cosh^{-1} \frac{a}{b}$$

Vertical Displacement at (0, y):

$$v(0, y) = \frac{2P}{\pi E'} \left(\tanh^{-1} \sqrt{\frac{a^2 - b^2}{a^2 + y^2}} + \alpha \frac{y^2}{b^2 + y^2} \sqrt{\frac{a^2 - b^2}{a^2 + y^2}} \right)$$

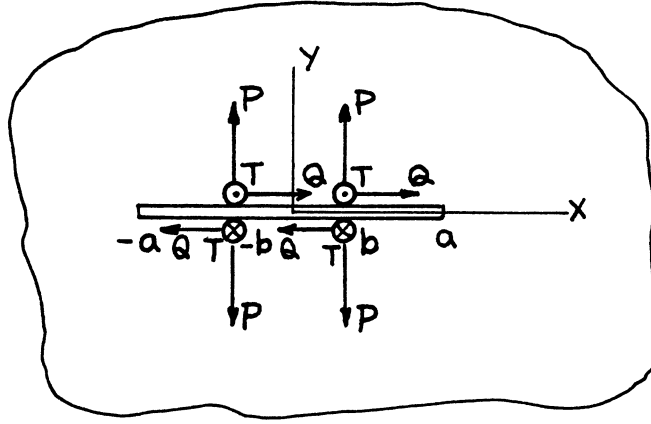
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Method: A , v Paris' Equation (see **Appendix B**) or Reciprocity (see **page 5.8b** and **page 5.9**)

Accuracy: Exact

Reference: **Tada 1985**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{\sqrt{a^2 - b^2}}{(z^2 - b^2)\sqrt{1 - (a/z)^2}}$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \\ \bar{Z}_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \tan^{-1} \sqrt{\frac{z^2 - a^2}{a^2 - b^2}}$$

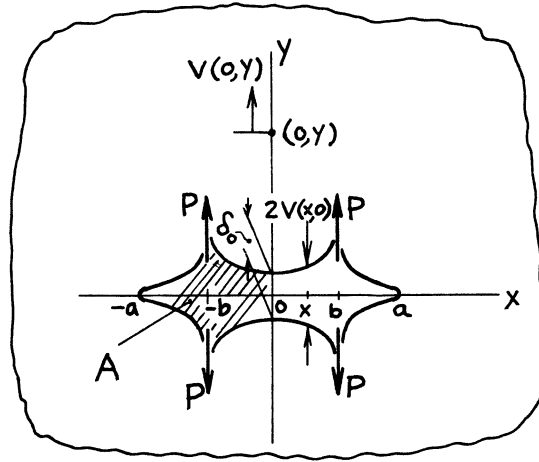
$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{a}{\sqrt{a^2 - b^2}}$$

$$\operatorname{Im} \begin{Bmatrix} \bar{Z}_I(x) \\ \bar{Z}_{II}(x) \\ \bar{Z}_{III}(x) \end{Bmatrix} \Big|_{|x| \leq a} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \ln \left| \frac{\sqrt{a^2 - x^2} + \sqrt{a^2 - b^2}}{\sqrt{a^2 - x^2} - \sqrt{a^2 - b^2}} \right|$$

Method: Westergaard Stress Function, etc. (or Superposition of **page 5.10**)

Accuracy: Exact

References: **Irwin 1957, 1958a; Erdogan 1962; Sih 1962a, 1964; Paris 1965**



$$K_I = \frac{2P}{\sqrt{\pi a}} \frac{1}{\sqrt{1 - (b/a)^2}}$$

Crack Opening Area:

$$A = \frac{8P}{E'} \sqrt{a^2 - b^2}$$

Crack Opening Profile:

$$2v(x, 0) = \frac{8P}{\pi E'} \left(\frac{\tanh^{-1}}{\coth^{-1}} \right) \sqrt{\frac{a^2 - b^2}{a^2 - x^2}} \quad \left(\begin{array}{l} a \leq |x| < b \\ |x| > b \end{array} \right)$$

Opening at Center:

$$\delta_0 = 2v(0, 0) = \frac{8P}{\pi E'} \cosh^{-1} \frac{a}{b}$$

Vertical Displacement at (0, y):

$$v(0, y) = \frac{4P}{\pi E'} \left(\tanh^{-1} \sqrt{\frac{a^2 - b^2}{a^2 + y^2}} + \alpha \frac{y^2}{b^2 + y^2} \sqrt{\frac{a^2 - b^2}{a^2 + y^2}} \right)$$

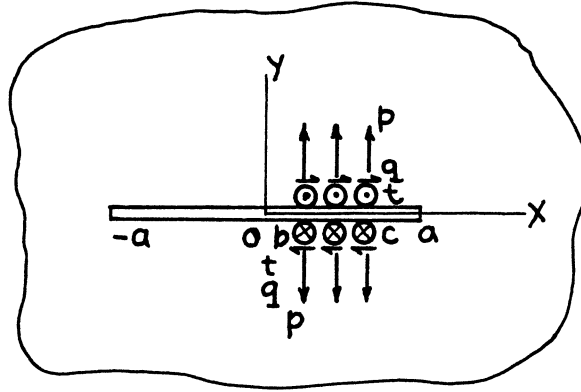
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Method: Superposition (see **pages 5.10 and 5.10a**)

Accuracy: Exact

Reference: **Tada 1985**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left[\sin^{-1} \frac{a^2 - cz}{a(z - c)} - \sin^{-1} \frac{a^2 - bz}{a(z - b)} + \frac{\sin^{-1} \frac{c}{a} - \sin^{-1} \frac{b}{a}}{\sqrt{1 - (a/z)^2}} - \frac{\sqrt{a^2 - c^2} - \sqrt{a^2 - b^2}}{\sqrt{z^2 - a^2}} \right]$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \\ \bar{Z}_{III}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left[(z - c) \sin^{-1} \frac{a^2 - cz}{a(z - c)} - (z - b) \sin^{-1} \frac{a^2 - bz}{a(z - b)} + \left(\sin^{-1} \frac{c}{a} - \sin^{-1} \frac{b}{a} \right) \sqrt{z^2 - a^2} \right]$$

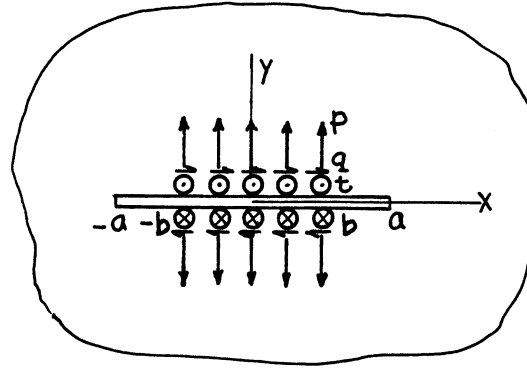
$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_{\pm a} = \frac{1}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \sqrt{\pi a} \left[\sin^{-1} \frac{c}{a} - \sin^{-1} \frac{b}{a} \mp \left(\sqrt{1 - (c/a)^2} - \sqrt{1 - (b/a)^2} \right) \right]$$

$$\text{Im} \begin{Bmatrix} \bar{Z}_I(x) \\ \bar{Z}_{II}(x) \\ \bar{Z}_{III}(x) \end{Bmatrix}_{|x| \leq a} = \frac{1}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left[(c - x) \cosh^{-1} \frac{a^2 - cx}{a|x - c|} - (b - x) \cosh^{-1} \frac{a^2 - bx}{a|x - b|} + \left(\sin^{-1} \frac{c}{a} - \sin^{-1} \frac{b}{a} \right) \sqrt{a^2 - x^2} \right]$$

Method: Integration of **page 5.10**

Accuracy: Exact

References: **Erdogan 1962; Sih 1962a, 1964; Paris 1965**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left[\frac{\sin^{-1} \frac{b}{a}}{\sqrt{1 - (a/z)^2}} - \tan^{-1} \sqrt{\frac{1 - (a/z)^2}{(a/b)^2 - 1}} \right]$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \\ \bar{Z}_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left[\left(\sin^{-1} \frac{b}{a} \right) \sqrt{z^2 - a^2} - z \tan^{-1} \sqrt{\frac{1 - (a/z)^2}{(a/b)^2 - 1}} + b \tan^{-1} \sqrt{\frac{(z/a)^2 - 1}{1 - (b/a)^2}} \right]$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left(\sin^{-1} \frac{b}{a} \right) \sqrt{\pi a}$$

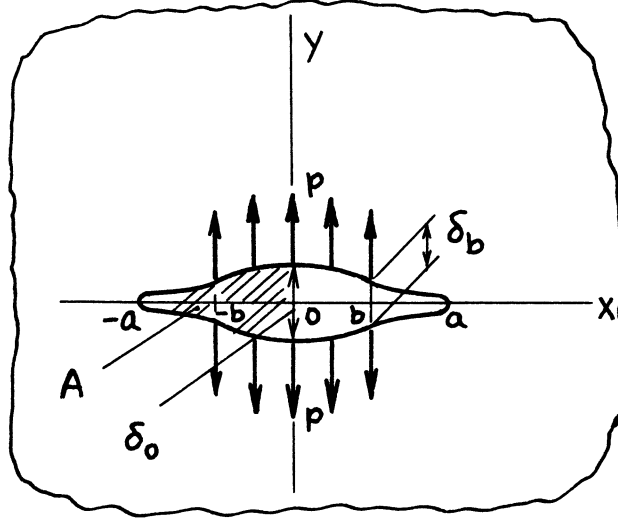
$$\begin{aligned} \text{Im} \begin{Bmatrix} \bar{Z}_I(x) \\ \bar{Z}_{II}(x) \\ \bar{Z}_{III}(x) \end{Bmatrix} \Big|_{|x| \leq a} &= \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left[\left(\sin^{-1} \frac{b}{a} \right) \sqrt{a^2 - x^2} - x \left(\frac{\coth^{-1}}{\tanh^{-1}} \right) \sqrt{\frac{(a/x)^2 - 1}{(a/b)^2 - 1}} \right. \\ &\quad \left. + b \left(\frac{\coth^{-1}}{\tanh^{-1}} \right) \sqrt{\frac{1 - (x/a)^2}{1 - (b/a)^2}} \right] \begin{pmatrix} |x| < b \\ b < |x| \leq a \end{pmatrix} \end{aligned}$$

$$\text{Im} \begin{Bmatrix} \bar{Z}_I(x) \\ \bar{Z}_{II}(x) \\ \bar{Z}_{III}(x) \end{Bmatrix} \Big|_{|x| = b} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left[\left(\sin^{-1} \frac{b}{a} \right) \sqrt{a^2 - b^2} + b \ln \frac{a}{b} \right]$$

Method: Special Case of **page 5.12**

Accuracy: Exact

References: **Erdogan 1962; Sih 1962a, 1964; Paris 1965**



$$K_I = p\sqrt{\pi a} \cdot \frac{2}{\pi} \sin^{-1} \frac{b}{a}$$

Crack Opening Area:

$$A = \frac{2pa^2}{E'} \left\{ \sin^{-1} \frac{b}{a} + \frac{b}{a} \sqrt{1 - \left(\frac{b}{a}\right)^2} \right\}$$

Opening at Center:

$$\delta_0 = 2v(0, 0) = \frac{8pa}{\pi E'} \left\{ \sin^{-1} \frac{b}{a} + \frac{b}{a} \cosh^{-1} \frac{a}{b} \right\}$$

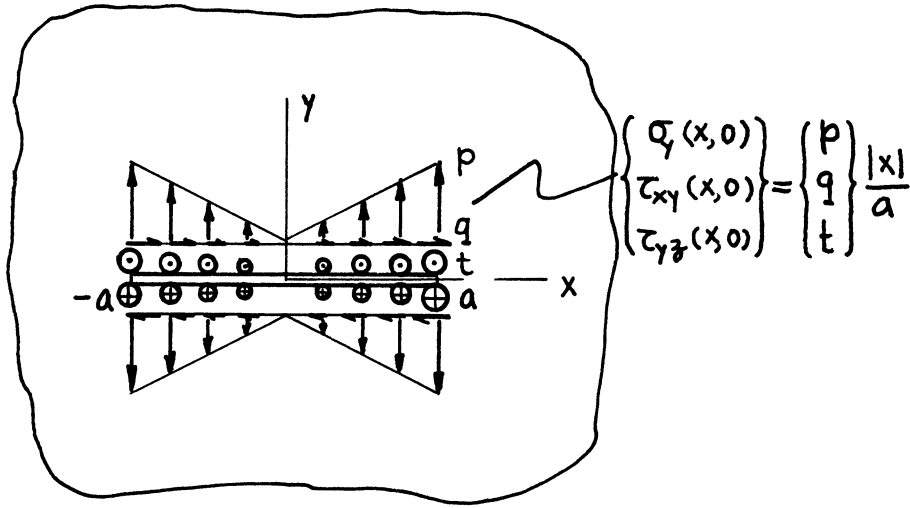
Opening at $x = b$:

$$\delta_b = 2v(\pm b, 0) = \frac{8pa}{\pi E'} \left\{ \sqrt{1 - \left(\frac{b}{a}\right)^2} \sin^{-1} \frac{b}{a} - \frac{b}{a} \ell n \frac{b}{a} \right\}$$

Method: A, δ Paris' Equation (see **Appendix B**) or Integration of **pages 5.11 and 5.11a**

Accuracy: Exact

Reference: **Tada 1985**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left\{ \frac{1}{\sqrt{1 - (a/z)^2}} + \frac{z}{a} \sin^{-1} \frac{a}{z} \right\}$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \\ \bar{Z}_{III}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left\{ \sqrt{z^2 - a^2} + \frac{z^2}{a} \sin^{-1} \frac{a}{z} \right\}$$

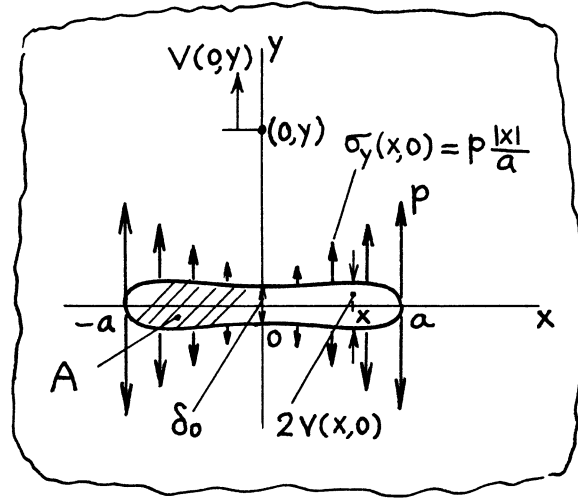
$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \sqrt{\pi a}$$

$$\operatorname{Im} \begin{Bmatrix} \bar{Z}_I(x) \\ \bar{Z}_{II}(x) \\ \bar{Z}_{III}(x) \end{Bmatrix} \Big|_{|x| \leq a} = \frac{1}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} a \left\{ \sqrt{1 - (x/a)^2} + \left(\frac{x}{a}\right)^2 \cosh^{-1} \frac{a}{x} \right\}$$

Method: Integration of **page 5.11**

Accuracy: Exact

Reference: **Tada 1973**



$$K_I = \frac{2}{\pi} p \sqrt{\pi a}$$

Crack Opening Area:

$$A = \frac{8}{3} \cdot \frac{pa^2}{E'}$$

Crack Opening Profile:

$$2v(x, 0) = \frac{4pa}{\pi E'} \left\{ \sqrt{1 - \left(\frac{x}{a}\right)^2} + \left(\frac{x}{a}\right)^2 \cosh^{-1} \frac{a}{x} \right\}$$

$|x| \leq a$

Opening at Center:

$$\delta_0 = 2v(0, 0) = \frac{4pa}{\pi E'}$$

Vertical Displacement at \$(0, y)\$:

$$v(0, y) = \frac{2p}{\pi E'} y \left\{ \sqrt{1 + \left(\frac{a}{y}\right)^2} - \frac{y}{a} \sinh^{-1} \frac{a}{y} - 2\alpha \left(\frac{1}{\sqrt{1 + \left(\frac{a}{y}\right)^2}} - \frac{y}{a} \sinh^{-1} \frac{a}{y} \right) \right\}$$

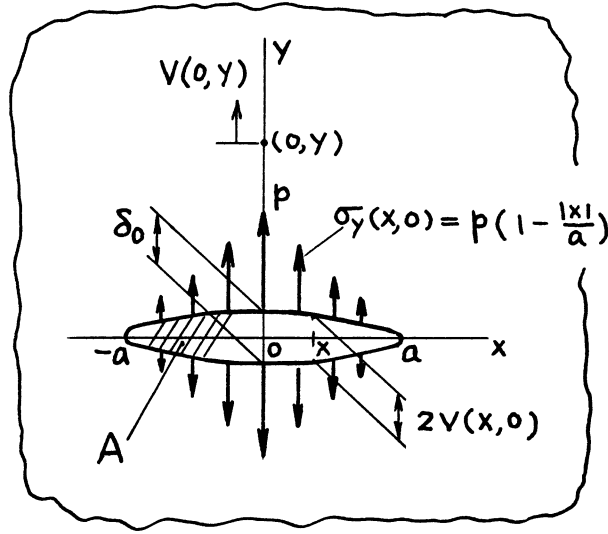
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Method: Integration of **pages 5.11 and 5.11a** or Paris' Equation (see **Appendix B**)

Accuracy: Exact

Reference: **Tada 1985**



$$K_I = \left(1 - \frac{2}{\pi}\right) p \sqrt{\pi a}$$

Crack Opening Area:

$$A = 2 \left(\pi - \frac{4}{3} \right) \frac{pa^2}{E'}$$

Crack Opening Profile:

$$2v(x,0) = \frac{4pa}{E'} \left\{ \left(1 - \frac{1}{\pi}\right) \sqrt{1 - \left(\frac{x}{a}\right)^2} - \frac{1}{\pi} \left(\frac{x}{a}\right)^2 \cosh^{-1} \frac{a}{x} \right\}$$

Opening at Center:

$$\delta_0 = 2v(0,0) = \frac{4pa}{E'} \left(1 - \frac{1}{\pi}\right)$$

Vertical Displacement at $(0,y)$:

$$v(0,y) = \frac{2p}{E'} \left(1 - \alpha y \frac{\partial}{\partial y}\right) \left\{ \left(1 - \frac{1}{\pi}\right) \sqrt{a^2 + y^2} - y \left(1 - \frac{1}{\pi} \frac{y}{a} \sinh^{-1} \frac{a}{y}\right) \right\}$$

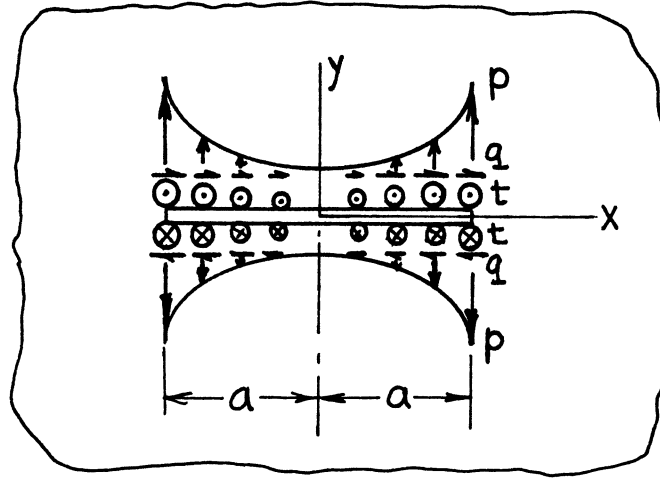
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Method: Superposition of **pages 5.1, 5.1a and 5.14, 5.14a**

Accuracy: Exact

Reference: **Tada 1985**



$$\begin{cases} \sigma_y(x, 0) \\ \tau_{xy}(x, 0) \\ \tau_{yz}(x, 0) \end{cases} = \begin{cases} p \\ q \\ t \end{cases} \left(\frac{|x|}{a} \right)^\gamma \quad (\gamma > -1) \\ |x| \leq a$$

$$\begin{cases} K_I \\ K_{II} \\ K_{III} \end{cases} = \begin{cases} p \\ q \\ t \end{cases} \sqrt{a} \frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2}+1\right)}$$

where

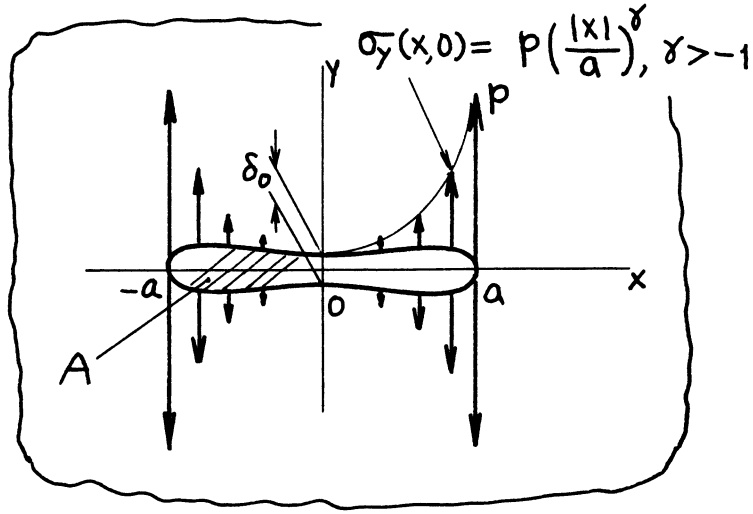
$\Gamma(\gamma)$ = Gamma Function (See **Appendix M**)

Methods: Fourier Transform (Sneddon); Integration of **page 5.11** (Tada)

Accuracy: Exact

References: **Sneddon 1951; Tada 1974**

NOTE: For special cases of $\gamma = 0$ and $\gamma = 1$, see **page 5.1** and **page 5.14**, respectively.



$$K_I = p\sqrt{a} \frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2}+1\right)}$$

Crack Opening Area:

$$A = \frac{2pa^2}{E'} \sqrt{\pi} \frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2}+2\right)}$$

Opening at Center:

$$\delta_0 = 2v(0,0) = \frac{4pa}{E'} \cdot \frac{1}{\sqrt{\pi}} \cdot \frac{1}{\gamma+1} \frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2}+1\right)}$$

where

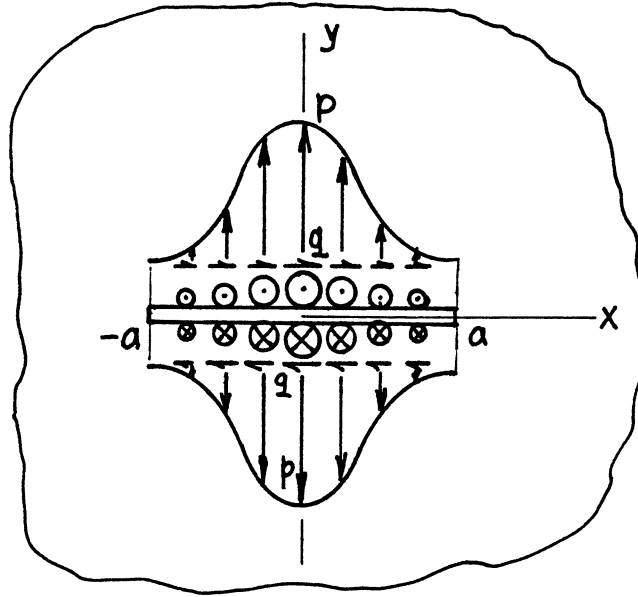
$\Gamma(\gamma)$ = Gamma Function (See **Appendix M**)

Method: Integration of **pages 5.11 and 5.11a** or Paris' Equation (see **Appendix B**)

Accuracy: Exact

Reference: **Tada 1985**

NOTE: For special cases $\gamma = 0$ and $\gamma = 1$, see **pages 5.1, 5.1a, and 5.14, 5.14a**.



$$\left\{ \begin{array}{l} \sigma_y(x, 0) \\ \tau_{xy}(x, 0) \\ \tau_{yz}(x, 0) \end{array} \right\}_{|x| \leq a} = \left\{ \begin{array}{l} p \\ q \\ t \end{array} \right\} \left\{ \sqrt{1 - \left(\frac{x}{a}\right)^2} \right\}^{\gamma} \quad (\gamma > -1)$$

$$\left\{ \begin{array}{l} K_I \\ K_{II} \\ K_{III} \end{array} \right\} = \left\{ \begin{array}{l} p \\ q \\ t \end{array} \right\} \sqrt{a} \frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2} + 1\right)}$$

where

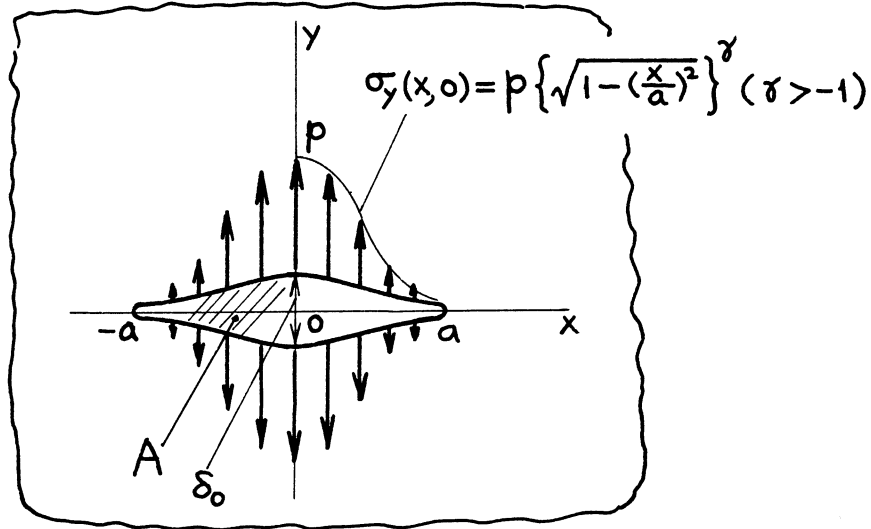
$\Gamma(\gamma)$ Gamma Function (See **Appendix M**)

Method: Integration of **page 5.11**

Accuracy: Exact

Reference: **Tada 1974**

NOTE: For special case of $\gamma = 0$, see **page 5.1**.



$$K_I = p\sqrt{a} \frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2} + 1\right)}$$

Crack Opening Area:

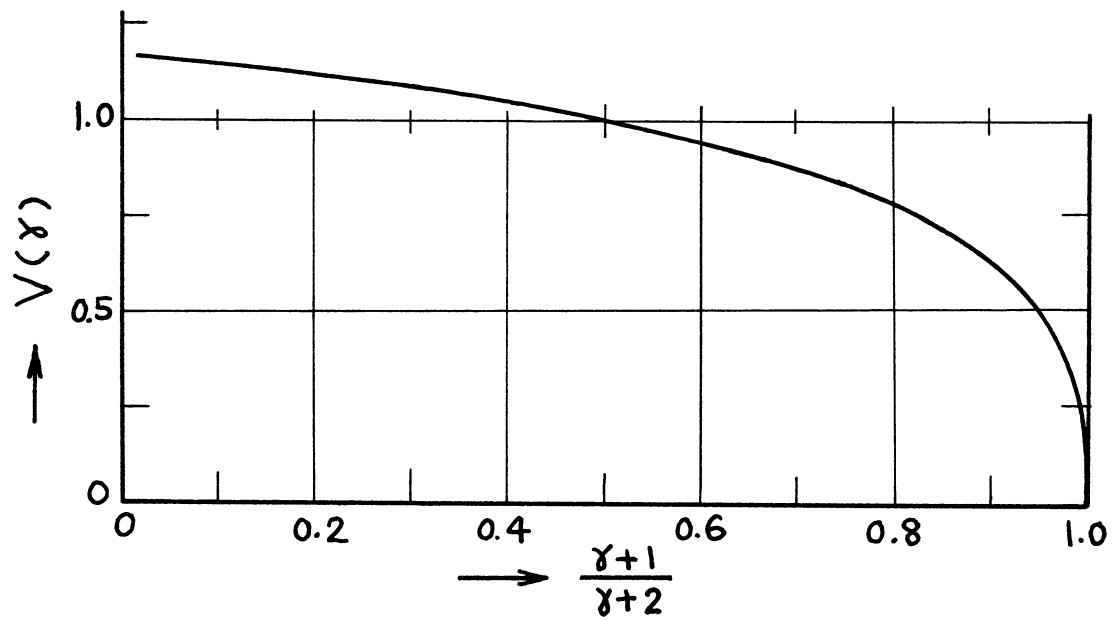
$$A = \frac{4pa^2}{E'} \sqrt{\pi} \frac{\Gamma\left(\frac{\gamma+3}{2}\right)}{\Gamma\left(\frac{\gamma}{2} + 2\right)}$$

Opening at Center:

$$\delta_0 = 2v(0,0) = \frac{4pa}{E'} V(\gamma)$$

where

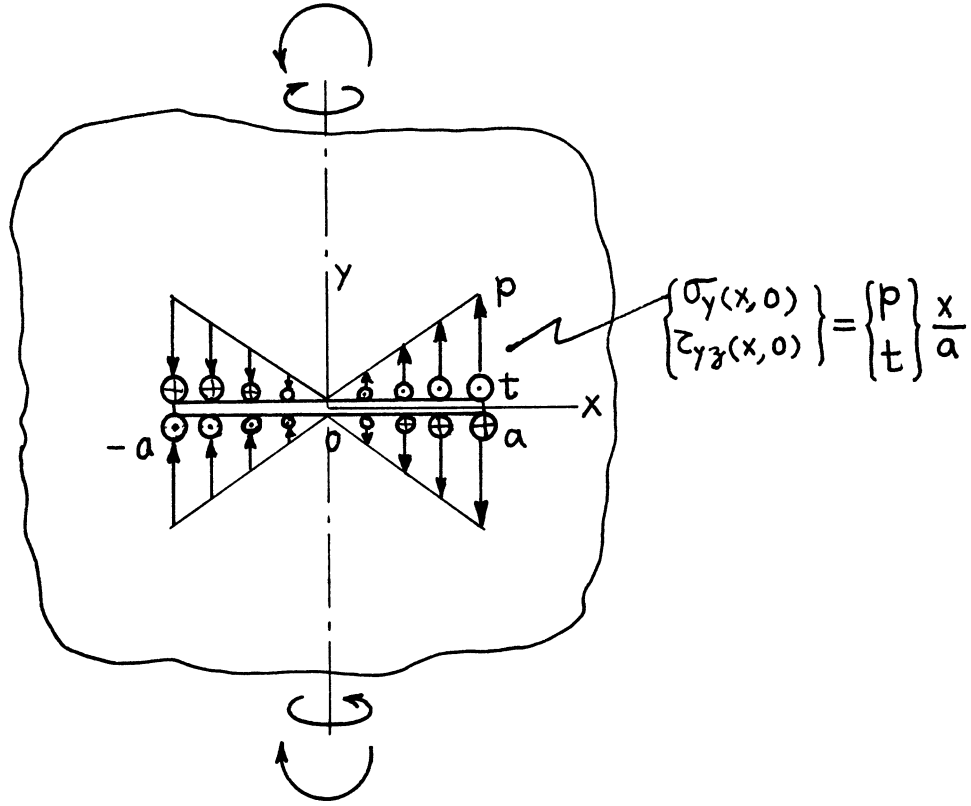
$\Gamma(\gamma)$ = Gamma Function (see **Appendix M**)



Method: Integration of **pages 5.11 and 5.11a**

Accuracy: A Exact; $V(\gamma)$ curve is based on accurate numerical values.

Reference: **Tada 1985**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{1}{2a} \begin{Bmatrix} p \\ t \end{Bmatrix} \frac{2z^2 - a^2}{\sqrt{z^2 - a^2}}$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{III}(z) \end{Bmatrix} = \frac{1}{2a} \begin{Bmatrix} p \\ t \end{Bmatrix} \left[z \sqrt{z^2 - a^2} \right]$$

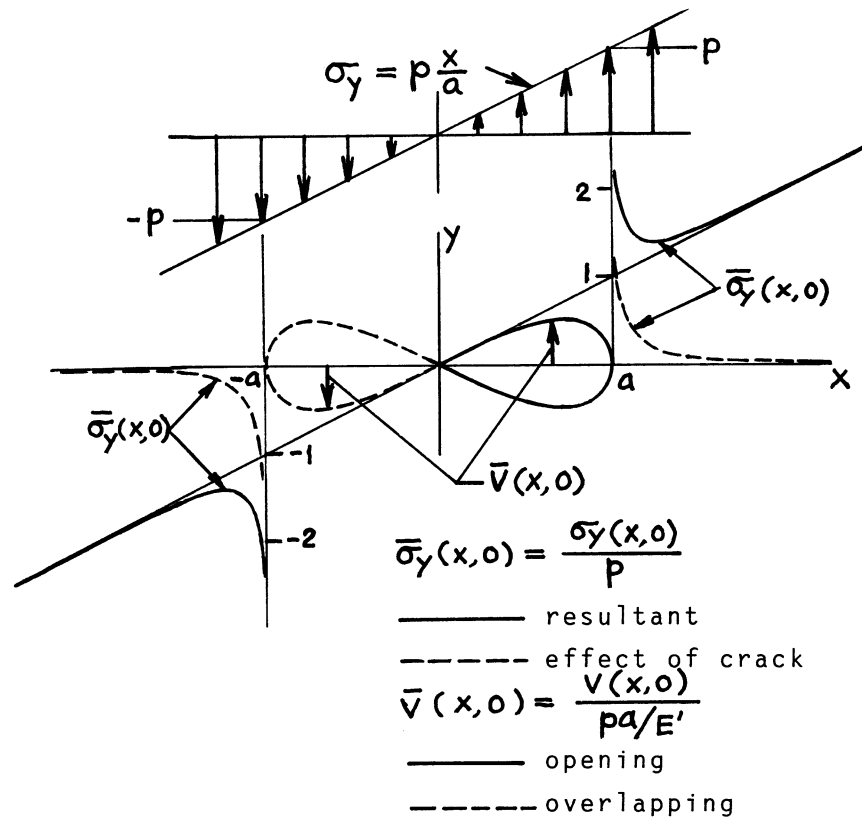
$$\begin{Bmatrix} K_I \\ K_{III} \end{Bmatrix}_{\pm a} = \pm \frac{1}{2} \begin{Bmatrix} p \\ t \end{Bmatrix} \sqrt{\pi a}$$

$$\operatorname{Im} \begin{Bmatrix} \bar{Z}_I(x) \\ \bar{Z}_{III}(x) \end{Bmatrix} \Big|_{|x| \leq a} = \frac{1}{2a} \begin{Bmatrix} p \\ t \end{Bmatrix} \left[x \sqrt{a^2 - x^2} \right]$$

Method: Integration of **page 5.11** or by Stress Concentration Factor

Accuracy: Exact

References: **Neuber 1937; Benthem 1972**



$$K_{I\pm a} = \pm \frac{1}{2} p \sqrt{\pi a}$$

$$\sigma_y(x,0) = p \frac{|x|}{x} \frac{\left(\frac{x}{a}\right)^2 - \frac{1}{2}}{\sqrt{\left(\frac{x}{a}\right)^2 - 1}}$$

$$2v(x,0) = \frac{2pa}{E'} \cdot \left(\frac{x}{a}\right) \sqrt{1 - \left(\frac{x}{a}\right)^2}$$

Crack Opening Area of Right Half:

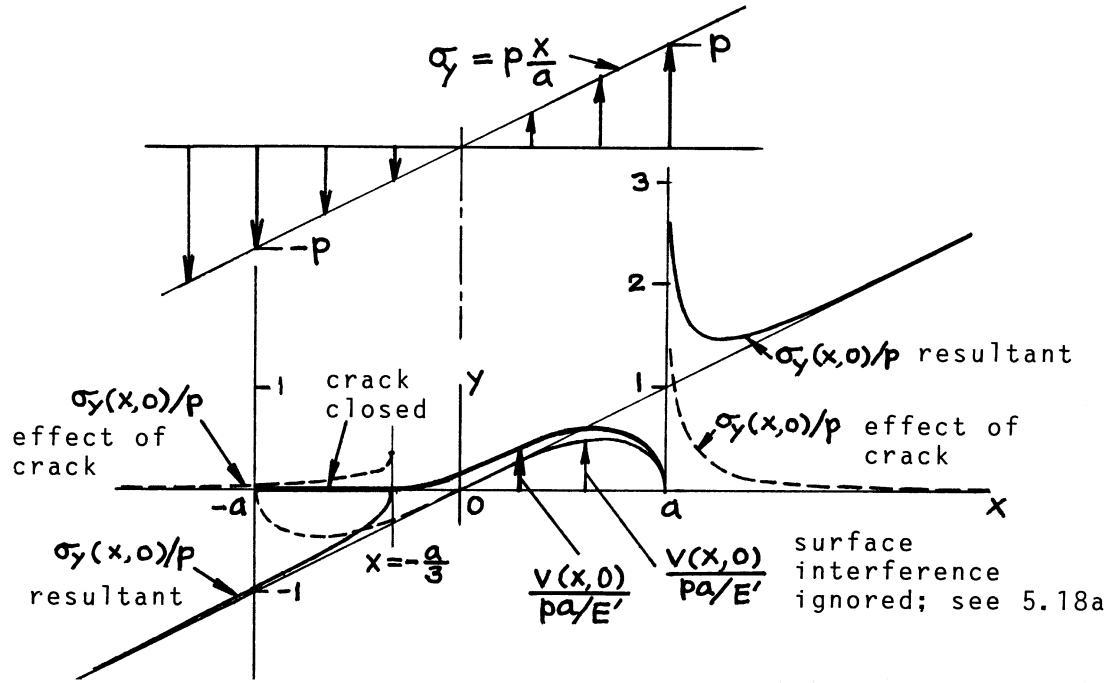
$$A_{x>0} = \frac{2pa^2}{3E'}$$

Method: Westergaard Stress Functions

Accuracy: Exact

Reference: **Tada 2000**

NOTE: Crack surface interference was ignored. See **page 5.18b** for the effect of surface interference.



Crack closes from $x = -a$ to $x = -a/3$.

$$Z_I(z) = \frac{p}{a} \left(z - \frac{2a}{3} \right) \sqrt{\frac{z + a/3}{z - a}}$$

$$\bar{Z}_I(z) = \frac{p}{2a} \left(z + \frac{a}{3} \right) \sqrt{(z + a/3)(z - a)}$$

$$K_{I, x=a} = \left(\frac{2}{3} \right)^{3/2} p \sqrt{\pi a} = 0.5443 p \sqrt{\pi a}; \quad K_{I, x=-a/3} = 0$$

$$\sigma_y(x, 0) = \frac{p}{a} \left(x - \frac{2a}{3} \right) \sqrt{\frac{x + a/3}{x - a}} \quad \text{for } x < -a/3, x > a$$

$$2v(x, 0) = \frac{2p}{E'} \left(x + \frac{a}{3} \right) \sqrt{\left(x + \frac{a}{3} \right) (a - x)} \quad \text{for } -a/3 \leq x \leq a$$

Crack Opening Area:

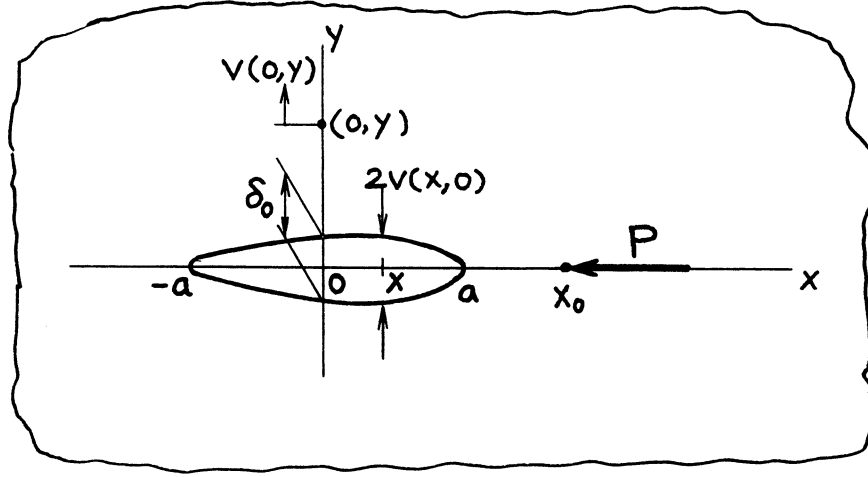
$$A = \left(\frac{2}{3}\right)^3 \frac{\pi p a^2}{E'} = 0.9308 \frac{p a^2}{E'}$$

Method: Superposition of **pages 5.1/1a and 5.18/18a**

Accuracy: Exact

References: **Seeger 1973; Tada 2000**

NOTE: Compare with **page 5.18a** for the effect of crack surface interference.



$$K_{I+a} = \frac{P}{2\sqrt{\pi a}} (1 - \alpha) \left(\sqrt{\frac{x_0 + a}{x_0 - a}} - 1 \right)$$

$$K_{I-a} = \frac{P}{2\sqrt{\pi a}} (1 - \alpha) \left(1 - \sqrt{\frac{x_0 - a}{x_0 + a}} \right)$$

Crack Opening Area:

$$A = \frac{2(1 - \alpha)}{E'} Pa \left\{ \frac{x_0}{a} - \sqrt{\left(\frac{x_0}{a} \right)^2 - 1} \right\}$$

Crack Opening Profile:

$$2v(x, 0) = \frac{2(1 - \alpha)}{\pi E'} P \left\{ \sin^{-1} \frac{1 - \frac{x_0 x}{a^2}}{\frac{x_0}{a} - \frac{x}{a}} + \sin^{-1} \frac{x}{a} \right\}$$

$|x| \leq a$

Opening at Center:

$$\delta_0 = 2v(0, 0) = \frac{2(1 - \alpha)}{\pi E'} P \sin^{-1} \frac{a}{x_0}$$

Vertical Displacement at $(0, y)$:

$$v(0, y) = \frac{1 - \alpha}{\pi E'} P \left(1 - \alpha y \frac{\partial}{\partial y} \right) \left\{ \cos^{-1} \sqrt{\frac{x_0^2 - a^2}{x_0^2 + y^2}} - \cos^{-1} \frac{x_0}{\sqrt{x_0^2 + y^2}} \right\}$$

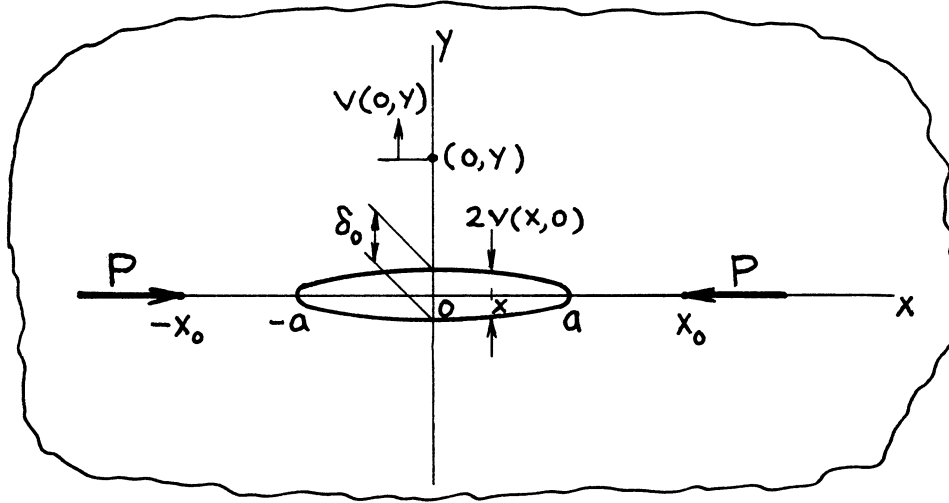
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Method: Integration of **pages 5.9, 5.9a**; A, δ_0 , One half of **page 5.20**

Accuracy: Exact

Reference: **Tada 1985**



$$K_I = \frac{P}{\sqrt{\pi a}} (1 - \alpha) \frac{a}{\sqrt{x_0^2 - a^2}}$$

Crack Opening Area:

$$A = \frac{4(1 - \alpha)}{E'} Pa \left\{ \frac{x_0}{a} - \sqrt{\left(\frac{x_0}{a}\right)^2 - 1} \right\}$$

Crack Opening Profile:

$$2v(x, 0) = \frac{4(1 - \alpha)}{\pi E'} P \cos^{-1} \sqrt{\frac{x_0^2 - a^2}{x_0^2 - x^2}} \quad |x| \leq a$$

Opening at Center:

$$\delta_0 = 2v(0, 0) = \frac{4(1 - \alpha)}{\pi E'} P \sin^{-1} \frac{a}{x_0}$$

Displacement at (0, y):

$$v(0, y) = \frac{2(1 - \alpha)}{\pi E'} P \left(1 - \alpha y \frac{\partial}{\partial y} \right) \left\{ \cos^{-1} \sqrt{\frac{x_0^2 - a^2}{x_0^2 + y^2}} - \cos^{-1} \frac{x_0}{\sqrt{x_0^2 + y^2}} \right\}$$

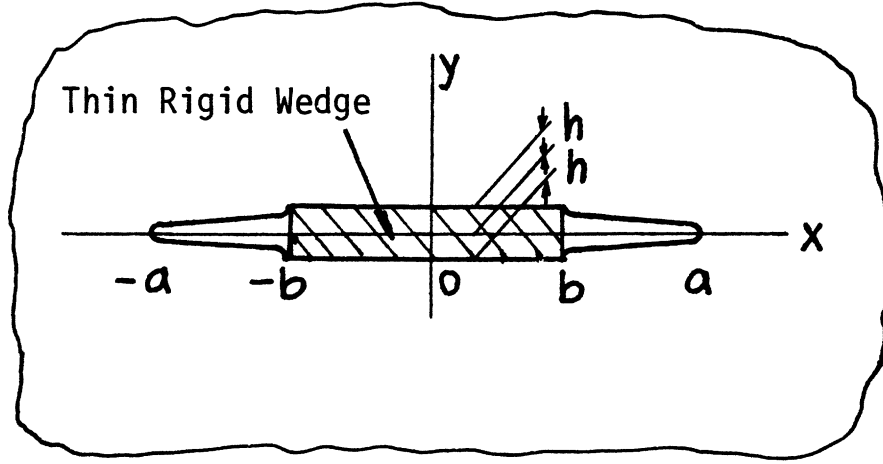
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Methods: Superposition of **page 5.19**; Paris' Equation for A , ν (see **Appendix B**)

Accuracy: Exact

Reference: **Tada 1985**



$$K_{I\pm a} = \frac{E'h}{2kK(k)} \sqrt{\frac{\pi}{a}}$$

$$K_{I\pm b} = -\frac{E'h}{2kK(k)} \sqrt{\frac{\pi}{b}}$$

$$2v(x,0) = 2h \cdot \frac{F(\varphi, k)}{K(k)} \quad b \leq |x| \leq a$$

where

$$E' = \begin{cases} E & \text{plane stress} \\ E/(1-\nu^2) & \text{plane strain} \end{cases}$$

$$k = \sqrt{1 - (b/a)^2}$$

$$\varphi = \sin^{-1} \sqrt{\frac{a^2 - x^2}{a^2 - b^2}}$$

$$F(\varphi, k) = \int_0^\varphi \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}$$

$$K(k) = F(\pi/2, k)$$

Methods: Muskhelishvili's Method (Markuzon); Triple Integral Equations (Tweed)

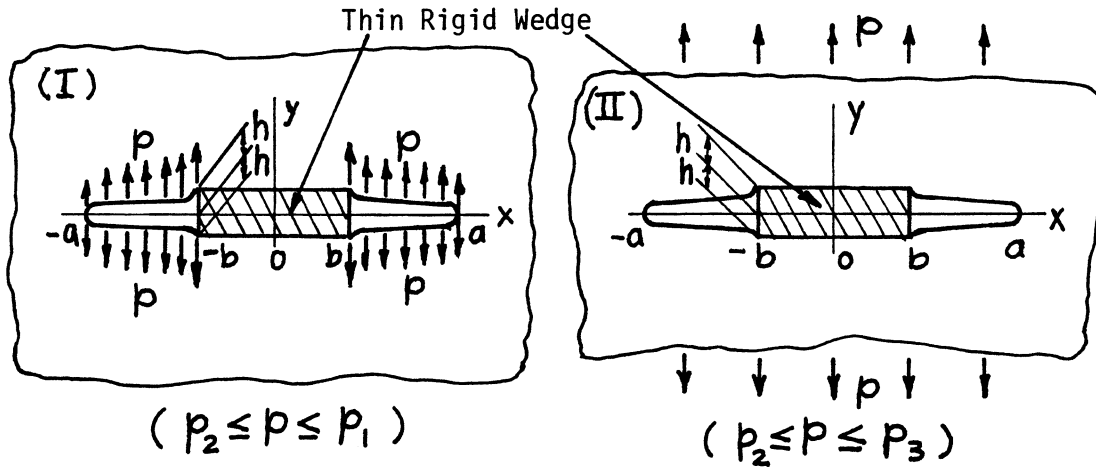
Accuracy: Exact

References: **Markuzon 1961; Tweed 1970; Tada 1974**

NOTE: For $K(k)$, see **Appendix L**.

The Westergaard Function is

$$Z_I(z) = \frac{E'ha}{2K(k)} \cdot \frac{1}{\sqrt{z^2 - a^2}} \cdot \frac{1}{\sqrt{z^2 - b^2}}$$



$$p_1 = \frac{E'ha}{2\{a^2E(k) - b^2K(k)\}}$$

$$p_2 = -\frac{E'h}{2a\{K(k) - E(k)\}}$$

$$p_3 = \frac{E'h}{2bK(k)}$$

$$K_I = \left\{ \frac{E'h}{2} + pa(K(k) - E(k)) \right\} \frac{1}{kK(k)} \sqrt{\frac{\pi}{a}}$$

$$K_I = \left\{ -\frac{E'h}{2} + p \frac{a^2E(k) - b^2K(k)}{a} \right\} \frac{1}{kK(k)} \sqrt{\frac{\pi}{b}}$$

$$2v(x, 0) = 2h \cdot \frac{F(\varphi, k)}{K(k)} + \frac{4pa}{E'} \{K(k)E(\varphi, k) - E(k)F(\varphi, k)\}$$

$b \leq |x| \leq a$

where

$$E' = \begin{cases} E & \text{plane stress} \\ E/(1 - \nu^2) & \text{plane strain} \end{cases}$$

$$k = \sqrt{1 - \left(b/a\right)^2}, \quad \varphi = \sin^{-1} \sqrt{\frac{a^2 - x^2}{a^2 - b^2}}$$

$$F(\varphi, k) = \int_0^\varphi \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}, \quad K(k) = F(\pi/2, k)$$

$$E(\varphi, k) = \int_0^\varphi \sqrt{1 - k^2 \sin^2 \varphi} d\varphi, \quad E(k) = E(\pi/2, k)$$

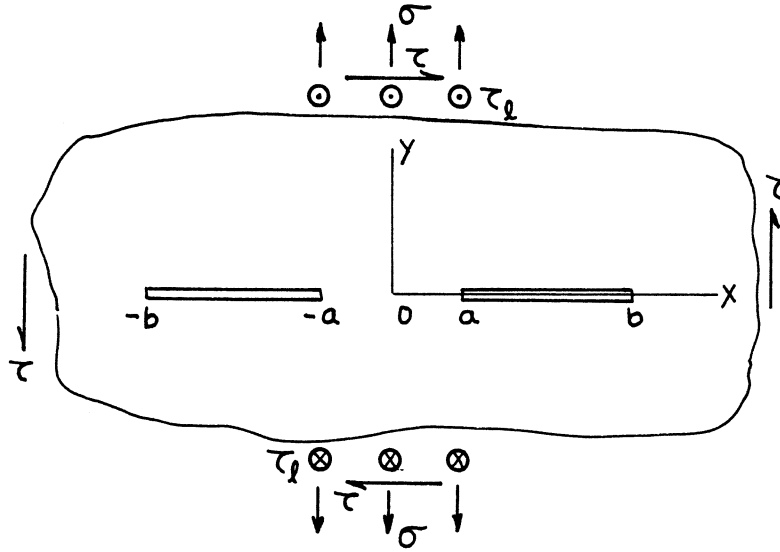
Methods: Triple Integral Equations or Superposition of **page 5.21** and **page 6.1**

Accuracy: Exact

References: **Tweed 1970; Tada 1974**

- NOTE:
1. In (I), when $p > p_1$, separation of contact surfaces occurs near $x = \pm b$, and when $p < p_2$, crack closure occurs near $x = \pm a$.
 2. In (II), when $p > p_3$, separation of contact surfaces occurs near $x = 0$, and when $p < p_2$, crack closure occurs near $x = \pm a$.
 3. For $K(k)$ and $E(k)$, see **Appendix L**

The Westergaard Function is $Z(z) = Z(z) + Z(z)$
p. 5.22 p. 5.21 p. 6.1



$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_{\pm a} = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \sqrt{\pi a} \frac{b^2 \frac{E(k)}{K(k)} - a^2}{a \sqrt{b^2 - a^2}}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_{\pm b} = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \sqrt{\pi b} \frac{1}{k} \left\{ 1 - \frac{E(k)}{K(k)} \right\}$$

where

$$k = \sqrt{1 - a^2/b^2}$$

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}$$

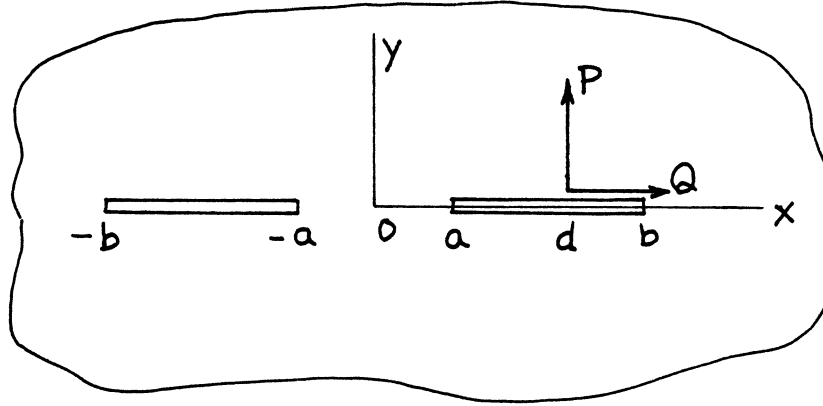
$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} d\varphi$$

(See **Appendix L** for values of $K(k)$ and $E(k)$)

Method: Muskhelishvili's Method

Accuracy: Exact

References: **Barenblatt 1962, Erdogan 1962, Sih 1964**



$$K_{I\pm b} = \frac{1}{2\sqrt{\pi b}\sqrt{b^2 - a^2}} \left\{ P\sqrt{d^2 - a^2} \sqrt{\frac{b\pm d}{b\mp d}} + \left(\frac{\kappa-1}{\kappa+1}\right) \left(1 \pm \frac{\pi}{2K(k)}\right) Qb \mp \frac{Pb}{K(k)} [E(k)F(\theta, k) - K(k)E(\theta, k)] \right\}$$

$$K_{II\pm b} = \frac{1}{2\sqrt{\pi b}\sqrt{b^2 - a^2}} \left\{ Q\sqrt{d^2 - a^2} \sqrt{\frac{b\pm d}{b\mp d}} - \left(\frac{\kappa-1}{\kappa+1}\right) \left(1 \pm \frac{\pi}{2K(k)}\right) Pb \mp \frac{Qb}{K(k)} [E(k)F(\theta, k) - K(k)E(\theta, k)] \right\}$$

$$K_{I\pm a} = \frac{1}{2\sqrt{\pi a}\sqrt{b^2 - a^2}} \left\{ P\sqrt{b^2 - d^2} \sqrt{\frac{d\pm a}{d\mp a}} - \left(\frac{\kappa-1}{\kappa+1}\right) Q \left(a \pm \frac{\pi}{2K(k)} b\right) \pm \frac{Pb}{K(k)} [E(k)F(\theta, k) - K(k)E(\theta, k)] \right\}$$

$$K_{II\pm a} = \frac{1}{2\sqrt{\pi a}\sqrt{b^2 - a^2}} \left\{ Q\sqrt{b^2 - d^2} \sqrt{\frac{d\pm a}{d\mp a}} + \left(\frac{\kappa-1}{\kappa+1}\right) P \left(a \pm \frac{\pi}{2K(k)} b\right) \pm \frac{Qb}{K(k)} [E(k)F(\theta, k) - K(k)E(\theta, k)] \right\}$$

where

$$k = \sqrt{1 - a^2/b^2}, \quad \theta = \sqrt{\frac{b^2 - d^2}{b^2 - a^2}}$$

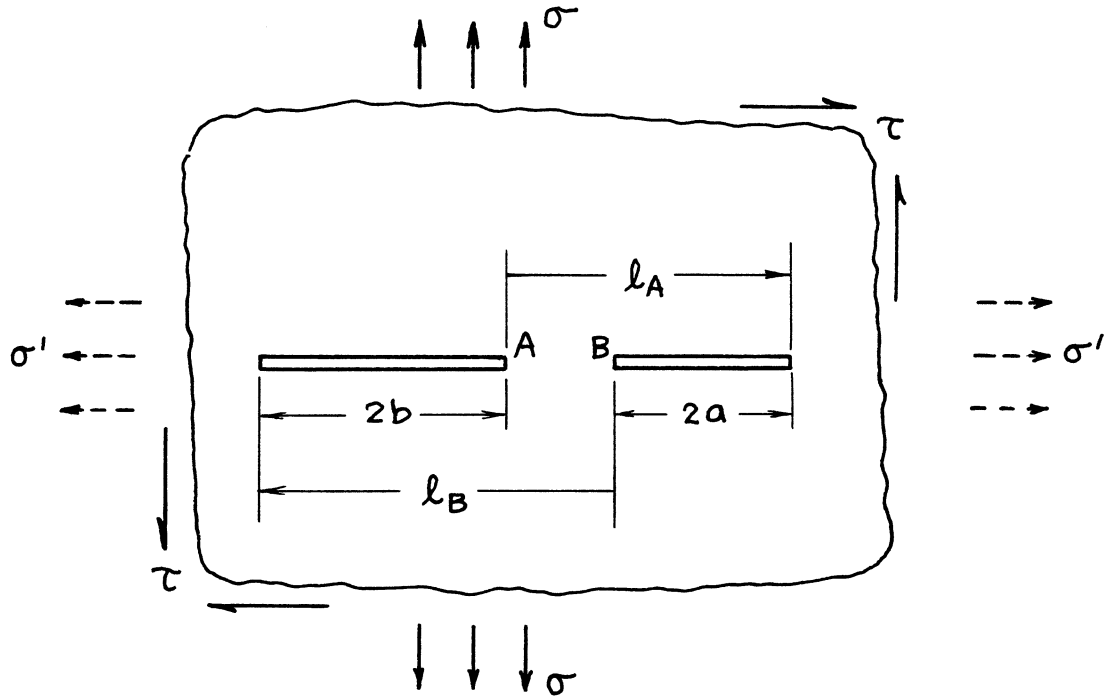
$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} \, d\varphi$$

$$F(\theta, k) = \int_0^\theta \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}, \quad E(\theta, k) = \int_0^\theta \sqrt{1 - k^2 \sin^2 \varphi} \, d\varphi$$

Method: Muskhelishvili's Method (See **Appendix L** for tables of $K(k)$ and $E(k)$)

Accuracy: Exact

References: **Barenblatt 1962, Erdogan 1962**



$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix}_A = \begin{Bmatrix} \sigma \\ \tau \end{Bmatrix} \sqrt{\pi b} \frac{1}{\sqrt{1 - \alpha_A}} \left[1 - \frac{1}{\alpha_B} \left\{ 1 - \frac{E(k)}{K(k)} \right\} \right]$$

$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix}_B = \begin{Bmatrix} \sigma \\ \tau \end{Bmatrix} \sqrt{\pi a} \frac{1}{\sqrt{1 - \alpha_B}} \left[1 - \frac{1}{\alpha_A} \left\{ 1 - \frac{E(k)}{K(k)} \right\} \right]$$

where

$$\alpha_A = \frac{2a}{\ell_A}, \quad \alpha_B = \frac{2b}{\ell_B}$$

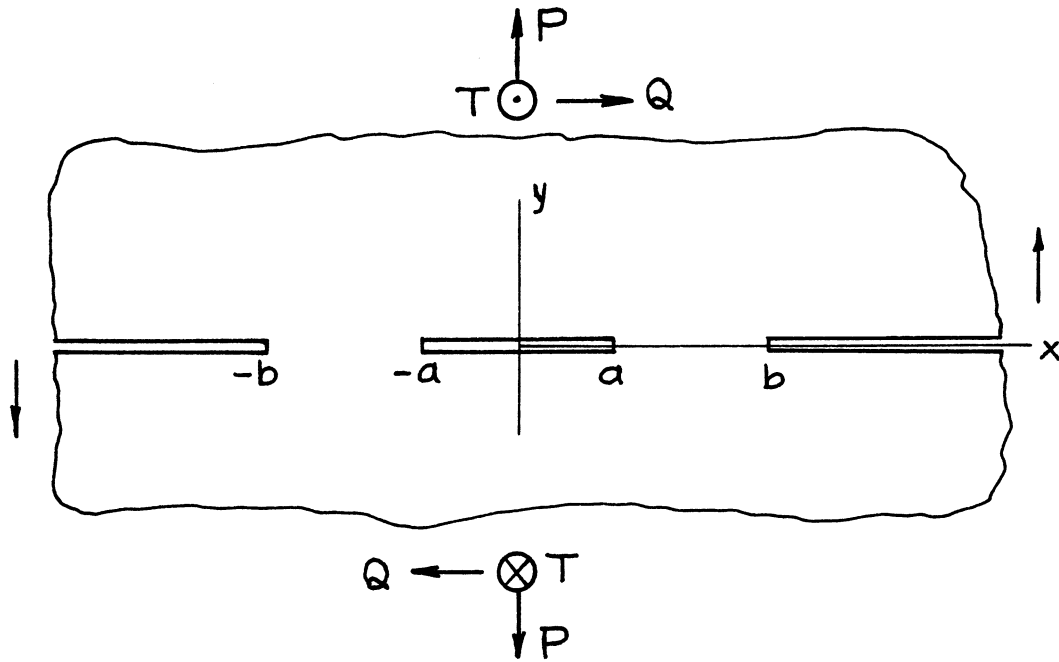
$$k = \sqrt{\alpha_A \alpha_B}$$

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} \, d\varphi$$

Method: Complex Potentials

Accuracy: Exact

References: Yokobori 1965 (or Kamei 1974), Isida 1973



$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_{\pm a} = \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{a}{\sqrt{b^2 - a^2}}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_{\pm b} = \frac{1}{\sqrt{\pi b}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{b}{\sqrt{b^2 - a^2}}$$

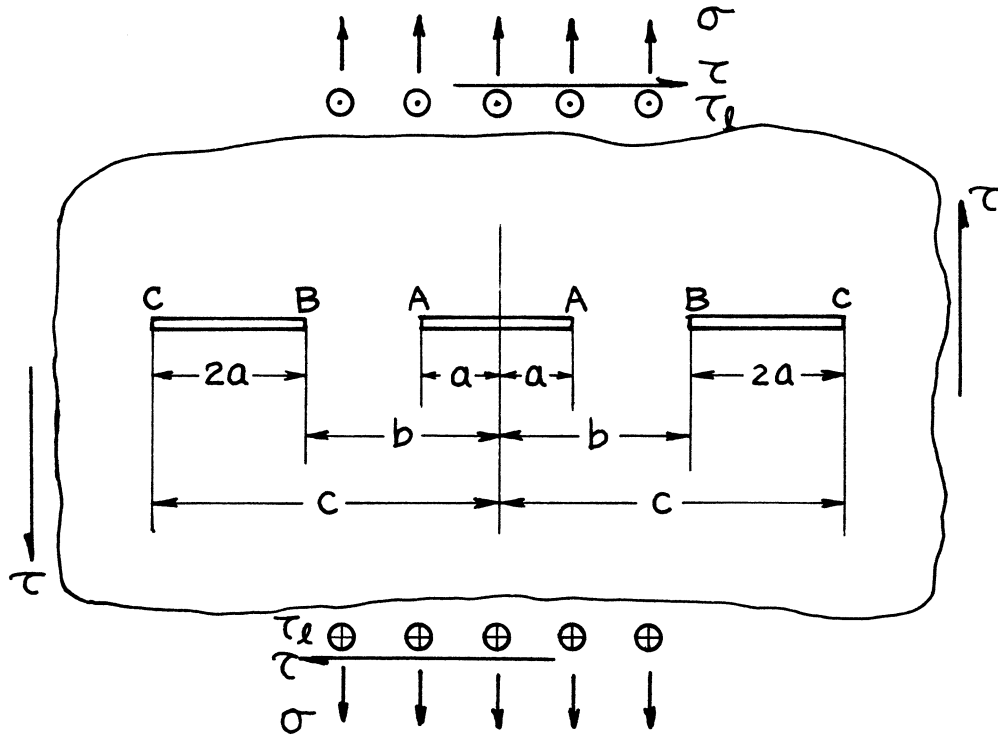
Methods: Westergaard Stress Function or from Solution of Punch Problem

Accuracy: Exact

References: **Galin 1953, Tada 1974**

The Westergaard Stress Function is

$$Z(z) = \frac{P}{\pi} \frac{z}{\sqrt{z^2 - a^2} \sqrt{b^2 - z^2}}$$



$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_A = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \sqrt{\pi a} \sqrt{\frac{c^2 - a^2}{b^2 - a^2}} \frac{E(k)}{K(k)}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_B = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \sqrt{\pi b} \sqrt{\frac{b^2 - a^2}{c^2 - b^2}} \left\{ \frac{c^2 - a^2}{b^2 - a^2} \frac{E(k)}{K(k)} - 1 \right\}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_C = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \sqrt{\pi c} \sqrt{\frac{c^2 - a^2}{c^2 - b^2}} \left\{ 1 - \frac{E(k)}{K(k)} \right\}$$

$$k^2 = (c^2 - b^2) / (c^2 - a^2)$$

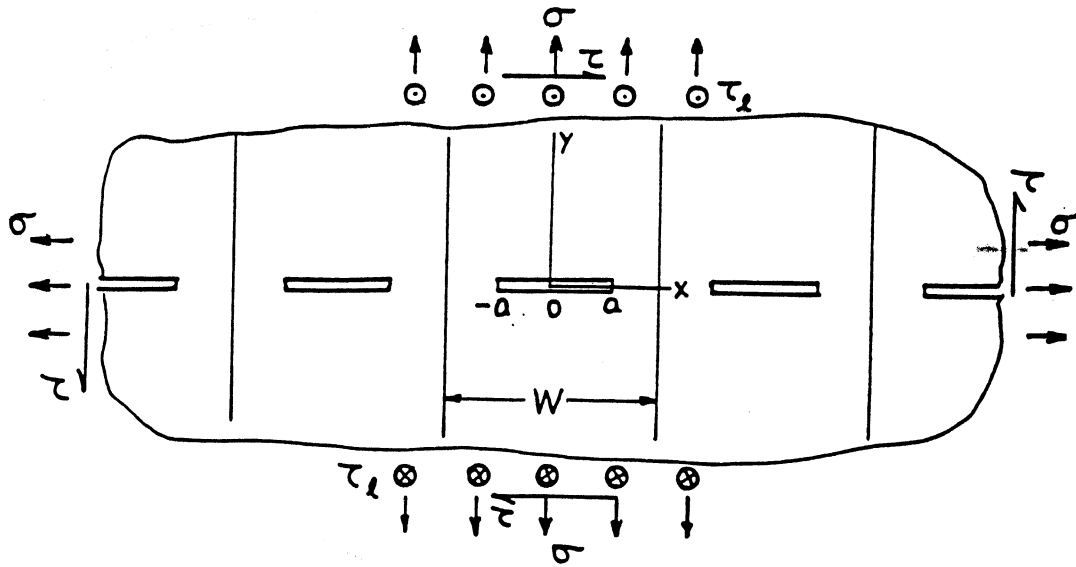
$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} \, d\varphi$$

(See **Appendix L** for values of $K(k)$ and $E(k)$.)

Method: Muskhelishvili's Method

Accuracy: Exact

Reference: Sih 1964



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \frac{1}{\sqrt{1 - \left(\sin \frac{\pi a}{W} / \sin \frac{\pi z}{W} \right)^2}}$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \\ \bar{Z}_{III}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} W \cos^{-1} \left(\frac{\cos \frac{\pi z}{W}}{\cos \frac{\pi a}{W}} \right)$$

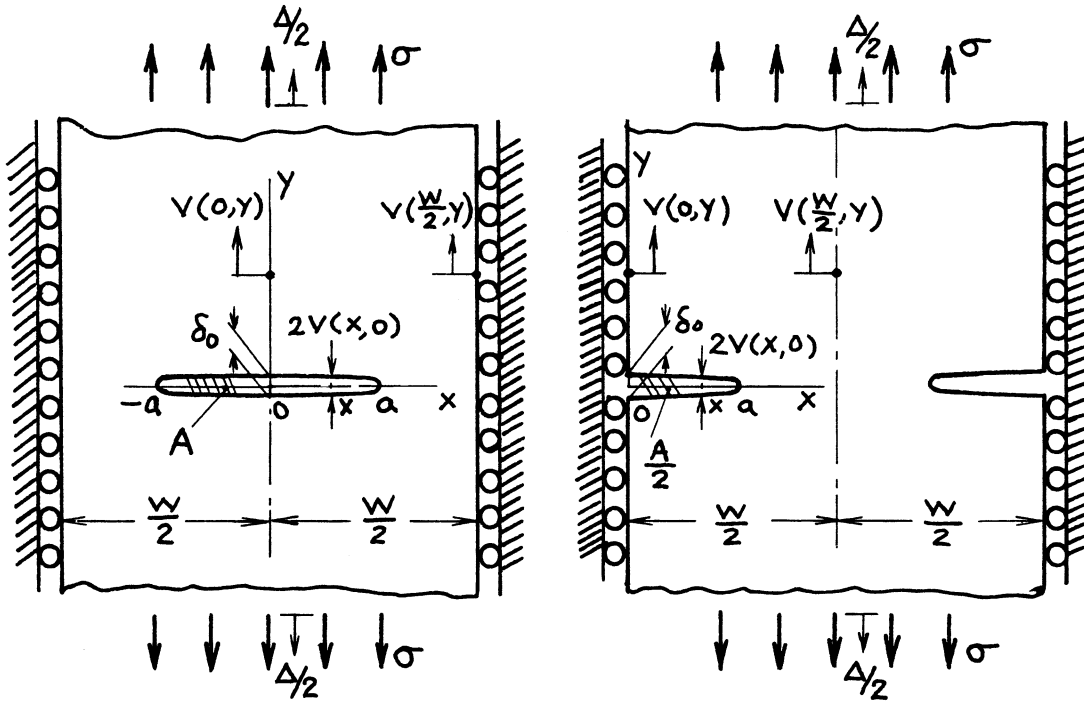
$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} \sqrt{W \tan \frac{\pi a}{W}}$$

$$\operatorname{Im} \begin{Bmatrix} \bar{Z}_I(x) \\ \bar{Z}_{II}(x) \\ \bar{Z}_{III}(x) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} \sigma \\ \tau \\ \tau_\ell \end{Bmatrix} W \cosh^{-1} \left(\frac{\cos \frac{\pi x}{W}}{\cos \frac{\pi a}{W}} \right) \quad |x| \leq a$$

Method: Westergaard Stress Function, etc. (Special Case of **page 7.9**)

Accuracy: Exact

References: **Irwin 1957; Koiter 1959**



$$K_I = \sigma \sqrt{W \tan \frac{\pi a}{W}}$$

Crack Opening Area:

$$A = \frac{4\sigma W^2}{\pi E'} \ell n \left(\sec \frac{\pi a}{W} \right)$$

Additional Relative Displacement at Infinity due to Crack:

$$\Delta_{crack} = A/W = \frac{4\sigma W}{\pi E'} \ell n \left(\sec \frac{\pi a}{W} \right) \quad (\Delta = \Delta_{no \text{ crack}} + \Delta_{crack})$$

Crack Opening Profile:

$$2v(x, 0) = \frac{4\sigma W}{\pi E'} \cosh^{-1} \left(\frac{\cos \frac{\pi x}{W}}{\cos \frac{\pi a}{W}} \right) \quad |x| \leq a$$

Opening at Center or Edge:

$$\delta_0 = 2v(0, 0) = \frac{4\sigma W}{\pi E'} \cosh^{-1} \left(\sec \frac{\pi a}{W} \right)$$

Additional Vertical Displacements at $(0, y)$ and $(W/2, y)$ due to Crack:

$$v(0, y) = \frac{2\sigma W}{\pi E'} \left(1 - \alpha y \frac{\partial}{\partial y} \right) \left\{ \cosh^{-1} \left(\frac{\cosh \frac{\pi y}{W}}{\cos \frac{\pi a}{W}} \right) - \frac{\pi y}{W} \right\}$$

$$v\left(\frac{W}{2}, y\right) = \frac{2\sigma W}{\pi E'} \left(1 - \alpha y \frac{\partial}{\partial y} \right) \left\{ \sinh^{-1} \left(\frac{\sinh \frac{\pi y}{W}}{\cos \frac{\pi a}{W}} \right) - \frac{\pi y}{W} \right\}$$

where

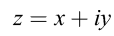
$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Methods: Westergaard Stress Function, Paris' Equation (see **Appendix B**)

Accuracy: Exact

Reference: **Tada 1985**

NOTE: $\Delta_{crack}/2$, $v(0, y)$ and $v(W/2, y)$ are the vertical displacements at $y = \infty$, $(0, y)$ and $(W/2, y)$, respectively, when uniform pressure σ is applied on the crack surfaces.

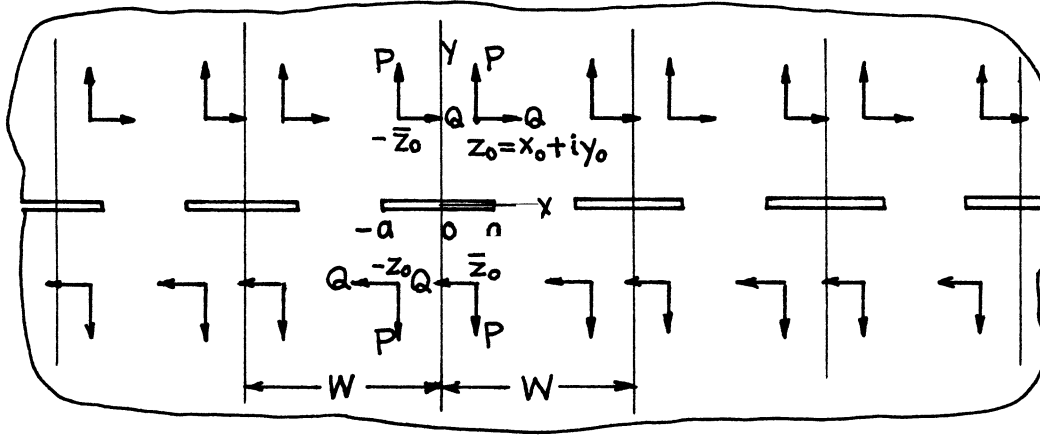


$$\left\{ \begin{array}{l} Z_I(z) \\ Z_{II}(z) \end{array} \right\} = \frac{1}{2W} \left\{ \begin{array}{l} P \\ Q \end{array} \right\} \left[1 \left\{ \begin{array}{l} - \\ + \end{array} \right\} \alpha y_0 \frac{\partial}{\partial y_0} \right] \left[\left\{ \begin{array}{l} \frac{\cos \frac{\pi z_0}{W} \sqrt{\left(\sin \frac{\pi a}{W} \right)^2 - \left(\sin \frac{\pi z_0}{W} \right)^2} + \frac{\cos \frac{\pi \bar{z}_0}{W} \sqrt{\left(\sin \frac{\pi a}{W} \right)^2 - \left(\sin \frac{\pi \bar{z}_0}{W} \right)^2}}{\sin \frac{\pi z}{W} - \sin \frac{\pi z_0}{W}} \\ \frac{\cos \frac{\pi z_0}{W} \sqrt{\left(\sin \frac{\pi a}{W} \right)^2 - \left(\sin \frac{\pi z_0}{W} \right)^2} + \frac{\cos \frac{\pi \bar{z}_0}{W} \sqrt{\left(\sin \frac{\pi a}{W} \right)^2 - \left(\sin \frac{\pi \bar{z}_0}{W} \right)^2}}{\sin \frac{\pi z}{W} - \sin \frac{\pi \bar{z}_0}{W}} \end{array} \right\} \right]$$

$$\left\{ \begin{array}{c} K_I \\ K_{II} \end{array} \right\}_{\pm a} = \frac{1}{2W} \left\{ \begin{array}{c} P \\ Q \end{array} \right\} \frac{\sqrt{W \tan \frac{\pi a}{W}}}{\sin \frac{\pi a}{W}} \left[1 \left\{ \begin{array}{c} - \\ + \end{array} \right\} \alpha_{y_0} \frac{\partial}{\partial y_0} \right] \left\{ \cos \frac{\pi z_0}{W} \sqrt{\frac{\sin \frac{\pi a}{W} \pm \sin \frac{\pi z_0}{W}}{\sin \frac{\pi a}{W} \mp \sin \frac{\pi z_0}{W}}} \right. \\ \left. + \cos \frac{\pi \bar{z}_0}{W} \sqrt{\frac{\sin \frac{\pi a}{W} \pm \sin \frac{\pi \bar{z}_0}{W}}{\sin \frac{\pi a}{W} \mp \sin \frac{\pi \bar{z}_0}{W}}} + i \left(\cos \frac{\pi z_0}{W} - \cos \frac{\pi \bar{z}_0}{W} \right) \right\}$$

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

References: Tada 1972a, 1973



$$z = x + iy$$

$$\begin{aligned} \begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \end{Bmatrix} &= \frac{1}{W} \begin{Bmatrix} P \\ Q \end{Bmatrix} \begin{bmatrix} 1 \\ + \end{bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \left[\left\{ \frac{\cos \frac{\pi z_0}{W} \sqrt{\left(\sin \frac{\pi a}{W} \right)^2 - \left(\sin \frac{\pi z_0}{W} \right)^2}}{\left(\sin \frac{\pi z}{W} \right)^2 - \left(\sin \frac{\pi z_0}{W} \right)^2} \right. \right. \\ &\quad \left. \left. + \frac{\cos \frac{\pi \bar{z}_0}{W} \sqrt{\left(\sin \frac{\pi a}{W} \right)^2 - \left(\sin \frac{\pi \bar{z}_0}{W} \right)^2}}{\left(\sin \frac{\pi z}{W} \right)^2 - \left(\sin \frac{\pi \bar{z}_0}{W} \right)^2} \right\} \frac{1}{\sqrt{\left(\sin \frac{\pi z}{W} \right)^2 - \left(\sin \frac{\pi a}{W} \right)^2}} \right] \end{aligned}$$

$$\begin{aligned} \begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \end{Bmatrix} &= \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \begin{bmatrix} 1 \\ + \end{bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \left[-\tan^{-1} \sqrt{\frac{\left(\cos \frac{\pi a}{W} / \cos \frac{\pi z}{W} \right)^2 - 1}{1 - \left(\cos \frac{\pi a}{W} / \cos \frac{\pi z_0}{W} \right)^2}} - \tan^{-1} \sqrt{\frac{\left(\cos \frac{\pi a}{W} / \cos \frac{\pi \bar{z}}{W} \right)^2 - 1}{1 - \left(\cos \frac{\pi a}{W} / \cos \frac{\pi \bar{z}_0}{W} \right)^2}} \right] \end{aligned}$$

$$\begin{aligned} \begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} &= \frac{1}{W} \begin{Bmatrix} P \\ Q \end{Bmatrix} \sqrt{W \tan \frac{\pi a}{W}} \begin{bmatrix} 1 \\ + \end{bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \left[\left\{ \frac{\cos \frac{\pi z_0}{W}}{\sqrt{\left(\sin \frac{\pi a}{W} \right)^2 - \left(\sin \frac{\pi z_0}{W} \right)^2}} + \frac{\cos \frac{\pi \bar{z}_0}{W}}{\sqrt{\left(\sin \frac{\pi a}{W} \right)^2 - \left(\sin \frac{\pi \bar{z}_0}{W} \right)^2}} \right\} \right] \end{aligned}$$

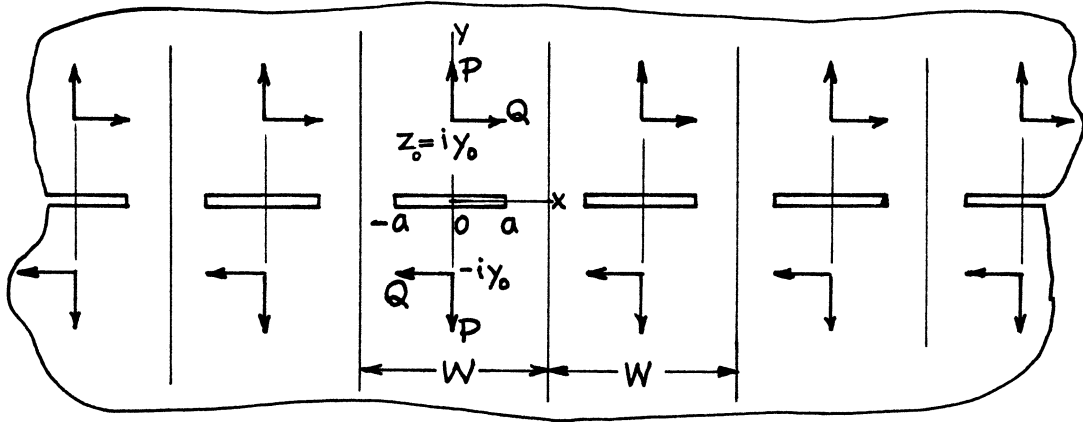
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function (Superposition of **page 7.2**)

Accuracy: Exact

References: **Tada 1972a, 1973**



$$z = x + iy$$

$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \end{Bmatrix} = \frac{1}{W} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \cosh \frac{\pi y_0}{W} \left[\frac{\sqrt{\left(\sin \frac{\pi a}{W} \right)^2 + \left(\sinh \frac{\pi y_0}{W} \right)^2}}{\left(\sin \frac{\pi z}{W} \right)^2 + \left(\sinh \frac{\pi y_0}{W} \right)^2} \frac{1}{\sqrt{1 - \left(\sin \frac{\pi a}{W} / \sin \frac{\pi z}{W} \right)^2}} \right]$$

$$\begin{Bmatrix} \bar{Z}_I(z) \\ \bar{Z}_{II}(z) \end{Bmatrix} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \left[-\tan^{-1} \frac{\left(\cos \frac{\pi a}{W} / \cos \frac{\pi z}{W} \right)^2 - 1}{\sqrt{1 - \left(\cos \frac{\pi a}{W} / \cosh \frac{\pi y_0}{W} \right)^2}} \right]$$

$$\begin{aligned} \begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} &= \frac{1}{W} \begin{Bmatrix} P \\ Q \end{Bmatrix} \sqrt{W \tan \frac{\pi a}{W}} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \frac{\cosh \frac{\pi y_0}{W}}{\sqrt{\left(\sin \frac{\pi a}{W} \right)^2 + \left(\sinh \frac{\pi y_0}{W} \right)^2}} \\ &= \frac{1}{W} \begin{Bmatrix} P \\ Q \end{Bmatrix} \sqrt{W \tan \frac{\pi a}{W}} \left[\frac{\cosh \frac{\pi y_0}{W}}{\sqrt{\left(\sin \frac{\pi a}{W} \right)^2 + \left(\sinh \frac{\pi y_0}{W} \right)^2}} \begin{Bmatrix} + \\ - \end{Bmatrix} \alpha \frac{\pi y_0}{W} \frac{\sinh \frac{\pi y_0}{W} \left(\cos \frac{\pi a}{W} \right)^2}{\left\{ \left(\sin \frac{\pi a}{W} \right)^2 + \left(\sinh \frac{\pi y_0}{W} \right)^2 \right\}^{3/2}} \right] \end{aligned}$$

$$\operatorname{Im} \begin{Bmatrix} \bar{Z}_I(x) \\ \bar{Z}_{II}(x) \end{Bmatrix}_{|x| \leq a} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \end{Bmatrix} \left[1 \begin{Bmatrix} - \\ + \end{Bmatrix} \alpha y_0 \frac{\partial}{\partial y_0} \right] \tanh^{-1} \frac{1 - \left(\cos \frac{\pi a}{W} / \cos \frac{\pi x}{W} \right)^2}{\sqrt{1 - \left(\cos \frac{\pi a}{W} / \cosh \frac{\pi y_0}{W} \right)^2}}$$

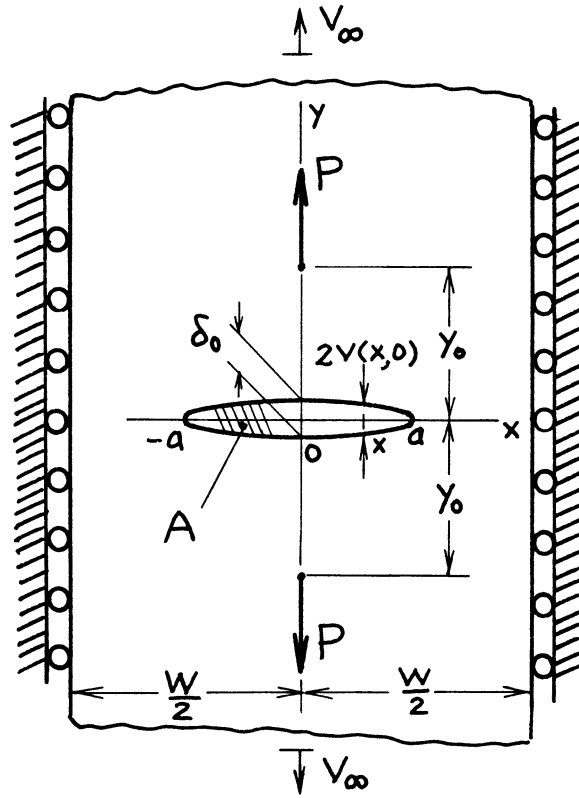
where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function (Special Case of **page 7.2 or 7.3**)

Accuracy: Exact

References: **Tada 1970, 1973**



$$K_I = \frac{P}{\sqrt{W}} \sqrt{\tan \frac{\pi a}{W} \left(1 - \alpha y_0 \frac{\partial}{\partial y_0}\right)} \frac{\cosh \frac{\pi y_0}{W}}{\sqrt{\left(\sin \frac{\pi a}{W}\right)^2 + \left(\sinh \frac{\pi y_0}{W}\right)^2}}$$

Crack Opening Area:

$$A = \frac{8PW}{\pi E'} \left(1 - \alpha y_0 \frac{\partial}{\partial y_0}\right) \left\{ \cosh^{-1} \left(\frac{\cosh \frac{\pi y_0}{W}}{\cos \frac{\pi a}{W}} \right) - \frac{\pi y_0}{W} \right\}$$

Relative Displacement at Infinity:

$$2v_\infty = A/W$$

Crack Opening Profile:

$$2v(x,0) = \frac{4P}{\pi E'} \left(1 - \alpha y_0 \frac{\partial}{\partial y_0}\right) \tanh^{-1} \sqrt{\frac{1 - \left(\cos \frac{\pi a}{W} / \cos \frac{\pi x}{W}\right)^2}{1 - \left(\cos \frac{\pi a}{W} / \cosh \frac{\pi y_0}{W}\right)^2}} \quad |x| \leq a$$

Crack Opening at Center:

$$\delta_0 = 2v(0, 0) = \frac{4P}{\pi E'} \left(1 - \alpha y_0 \frac{\partial}{\partial y_0} \right) \tanh^{-1} \left\{ \frac{\sin \frac{\pi a}{W}}{\sqrt{1 - \left(\cos \frac{\pi a}{W} / \cosh \frac{\pi y_0}{W} \right)^2}} \right\}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Methods: Westergaard Stress Function, Reciprocity (see **page 7.1a**)

Accuracy: Exact

Reference: **Tada 1985**

where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

Method: Westergaard Stress Function (Special Case of **page 7.3**)

Accuracy: Exact

References: **Tada 1970, 1973**